

Regional Needs Assessment and Capacity Strengthening for Vector Surveillance and Control in Central America and the Dominican Republic

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I. Background

Vector-borne diseases (VBD) remain a significant and persistent threat to global public health, accounting for approximately 17% of all infectious diseases and causing an estimated 700,000 deaths annually (WHO, 2024). The burden is particularly pronounced in tropical and subtropical regions, where over 80% of the global population is at risk from at least one major VBD, and more than half are exposed to two or more (WHO 2017a).

In recent years, the exponential rise in arboviral diseases in Latin America has intensified, emerging as a critical region on the global epidemic landscape. The 2015 Zika epidemic, combined with the unprecedented rise in dengue cases, yellow fever outbreaks, the circulation of Mayaro virus, and the newly emerging *Oropouche* virus (OROV) (Barçante and Cherem 2025), with its high potential for widespread impact, exemplifies the region's growing vulnerability. This complex and evolving epidemiological landscape is exacerbated by environmental challenges that promote ideal conditions to proliferate and facilitate their geographic spread (UNPD, 2009; de Almeida et al. 2025). As a result, outbreaks are no longer confined to traditionally endemic regions. As of 2024, over 13 million dengue cases alone have been recorded across North, Central, and South America, as well as the Caribbean, with transmission levels remaining elevated into 2025 (CDC 2025), and the lack of coordinated international action, particularly in places like Central America, may result in global proportion-like spread situations for vector-borne diseases.

This foreshadowing is evident in the increased dengue cases in common areas, among travellers, or in some instances, declared outbreaks in the continental United States during 2024. For example, dengue cases have been reported as far north as Florida (91 cases), California (18), Texas (1), and Puerto Rico (thousands of cases during 2022-2023), with a public health emergency declared in March 2024 for the latter. The locally acquired cases have risen 84% compared to 2023, with 3,483 cases among U.S. travellers identified (CDCHAN, 2025), indicating an increasing trend. Furthermore, locally acquired *Plasmodium* infections have also been reported in Florida (7 cases), Texas (1 case), and Maryland (1 case) (Fairfax County Health 2023). Since 2023, the OROV virus has started causing large outbreaks in endemic and new areas of South America, adding cases in Central America and the Caribbean (CDC 2025). Ultimately, during 2024-2025, 109 OROV virus disease cases based on travel status were reported in seven states and territories of the United States (AAP 2025).

Central America and the Dominican Republic (SICA)—with a combined population of approximately 62 million people— are especially vulnerable due to their geographic role as a land bridge, favourable climate, and high ecological diversity, all of which facilitate vector dispersal across the continent. While the region has made important progress in controlling certain VBDs, such as a 94% reduction in Chagas disease transmission between 2000–2010, and malaria elimination milestone in Belize and El Salvador (PAHO, 2023, 2025a), the threat of resurgence remains, with sporadic surges in cases and the risk of re-establishment of transmission remaining a constant concern. In addition,

addressing social determinants of health through integrated, cross-sector efforts, such as involving municipalities, is essential for sustainable vector-borne disease prevention in the subregion (SE-COMISCA 2009).

According to the *Global vector control response 2017–2030 (GVCR) WHO*, systemic and structural challenges continue to hinder vector control and surveillance. Most national and subnational programs operate with limited entomological capacity and poor infrastructure. In many endemic countries, disease-specific vertical programs are not fully integrated, leading to duplicated efforts and competition for limited resources. The SICA region continues to confront these critical challenges. Recent assessments highlight persistent gaps in entomological surveillance and vector control, including shortages of trained personnel, inadequate funding, and insufficient equipment (WHO 2003; Shooler 2017; Loureço et al. 2019). Surveillance systems for diseases like dengue and malaria often remain fragmented, with limited interoperability and weak links between entomological and epidemiological data. Although there is a wide range of vector control tools (WHO 2016, CARPHA 2017), the lack of digital tools for real-time data capture and analysis hampers timely outbreak detection and response. Furthermore, these limitations are compounded by growing insecticide resistance, rapid urbanization, and suboptimal housing conditions, complicating even further vector proliferation and control efforts (Fournet et al. 2018).

Russell et al. (2020) conducted a comprehensive needs assessment of the capacity of National Malaria Control Programmes (NMCPs) in countries of Asia-Pacific (n=14), Africa (n=18), and the Americas (Guatemala, Haiti, Honduras) for Malaria in efforts to identify strengths and weaknesses. Despite the magnitude of these challenges, no comprehensive, systematic needs assessment has yet been conducted to evaluate vector surveillance and control capacities across Central America and the Dominican Republic as a region. Available assessments are typically national in scope, treating countries as isolated systems rather than as interconnected parts of a shared epidemiological ecosystem. This fragmented approach limits the ability to prioritize regional investments, align strategies, and build sustainable, cooperative responses.

The Grupo Técnico de Entomología (GTEN)—under the coordination of SE-COMISCA and supported by partners including PAHO, CHAI, CDC, and UVG—brought together public health entomologists from across the SICA region facilitating exchanges in face of the persistent public health challenges from mosquito-borne diseases. The meeting held in April 2024 facilitated the exchange of experiences and identified shared challenges in areas such as vector surveillance, insecticide resistance monitoring, and vector control operationalization. Key barriers discussed included: absence of standardized insecticide rotation protocols, lack of integration between research and field operations, inefficiencies in insecticide procurement, outdated performance indicators, limited academia-program collaboration, and regulatory restrictions (GTEN 2024; PAHO 2024).

Recent documentation, including the April 2024 GTEN meeting report, highlights a consensus on critical barriers to effective vector control. These include the absence of systematic insecticide rotation protocols, insufficient integration of research into

operational practice, inefficiencies in insecticide procurement, outdated or inadequate prevention indicators, limited partnerships between academic institutions and vector control programs, and restrictive regulatory frameworks for insecticide registration. The GTEN has proposed a series of regional strategies to address these challenges, such as the development of regional guidelines for insecticide management, establishment of resistance reference centres, expert-driven indicator review panels, regulatory reforms, and digital monitoring platforms (GTEN 2024).

While the GTEN report is a valuable and critical reference that captures a wealth of regional expertise, the current state of vector surveillance and control, and consensus on priority areas for action in the SICA region, it does not constitute a formal, comprehensive needs assessment. Instead, it reflects the collective experience and perspectives of public health professionals from across the region rather than a structured evaluation of current capacities, needs, and gaps. This makes it an important reference point for understanding the current vector surveillance and control landscape, but it does not replace the need for a systematic, evidence-based Needs Assessment. It is essential to quantify resource deficiencies, map existing technical capacities, identify country-specific needs, and establish evidence-based priorities for investment and intervention in the region (GTEN 2024; PAHO 2024).

The lack of a recently conducted, region-wide Needs Assessment justifies the initiation of a formal process to guide future public health action. To date, the few assessments of vector surveillance and control capacities in Central America and the Dominican Republic have been conducted at the national level, treating countries as independent entities rather than as part of a connected regional system (PAHO 2019).

This evaluation is particularly necessary and timely, as it will represent the first comprehensive regional assessment of the current realities, barriers, and capacities of public health entomology laboratories and vector control programs in the SICA region. Given that vector-borne threats transcend borders, a regional, coordinated approach is not only more efficient—it is essential. Strengthening entomological and vector control capacities through a comprehensive needs assessment will enhance regional preparedness and response, mitigate disease risk, and ultimately safeguard public health (van der Berg et. al 2024) across Central America and the Dominican Republic.

The primary objective of this evaluation is to identify gaps in the capacities of public health entomology laboratories across the region, particularly in their ability to implement innovative entomological surveillance systems. By pinpointing these gaps, the assessment aims to support the development and deployment of advanced surveillance tools that can enable more efficient, timely, and targeted vector control strategies. Ultimately, strengthening these capacities is critical to reducing the risk of both emerging and endemic vector-borne diseases spreading within and beyond the region.

This needs assessment will be conducted within the framework of the Central American and Dominican Republic Vector Surveillance and Control Network (EntoNet) as part of

the strategy and broader efforts to strengthen and expand regional capacities within VecNet, the primary network to which EntoNet is a member.

The assessment is expected to identify the specific strengths and weaknesses of each national program, establish comparative baselines across countries facing similar epidemiological challenges, and generate concrete evidence to inform data-driven decision-making at both technical and political levels. The findings should facilitate the adaptation of national programs to new and evolving challenges, including the geographic expansion of vectors, rising insecticide resistance, and the impacts of climate change.

Furthermore, the assessment aims to align national efforts with global strategies, such as the Integrated Vector Management (IVN) framework promoted by the World Health Organization, ensuring that implemented interventions are sustainable, cost-effective, and tailored to the specific context of each country (WHO, 2025). This assessment represents a milestone for standardizing the track of surveillance and control of mosquito-borne diseases for the region and serves as an individual tool for the countries.

II. Objectives

1. Assess the status of national vector surveillance and control programs in the seven Central American nations (Belize, Guatemala, El Salvador, Nicaragua, Honduras, Costa Rica, and Panama) and the Dominican Republic, focusing on mosquito-borne disease vectors.
2. Identify program capacities and gaps related to their effectiveness and resilience in adequately controlling populations of mosquito-borne diseases.
3. Provide actionable recommendations to strengthen vector control programs across the region.

III. Methodology

A. Assessment Design and Scope

This regional needs assessment aims to evaluate the capacities, gaps, and operational realities of national and subnational vector surveillance and control programs in eight countries: Belize, Guatemala, El Salvador, Honduras, Nicaragua, Costa Rica, Panama, and the Dominican Republic. A cross-sectional assessment will be conducted between July and September 2025, focusing on mosquito-borne diseases—particularly dengue, malaria, and other arboviruses—and is intended to generate actionable recommendations for strengthening entomological surveillance systems and vector control strategies across the region.

The assessment will be implemented in two sequential phases at the national and subnational levels. A census approach and purposive sampling will be used in each phase, respectively.

B. Assessment Instrument and Variables

The assessment tool is adapted from the online questionnaire developed by PacMOSSI (Pacific Mosquito Surveillance Strengthening for Impact). Led by James Cook University in Australia, PacMOSSI has supported Pacific Island countries since 2021 in the prevention and control of mosquito-borne diseases, conducting comprehensive needs assessments in 2021 and 2024. In addition to PacMOSSI, we also followed the guidelines of the technical document “Framework for a National Needs Assessment for Vector Control” published by the World Health Organization and UNICEF (WHO & UNICEF, 2017) for further considerations as needed.

For this assessment, the original instrument has been further refined to reflect on lessons learned from PacMOSSI (Murphy, personal communication) and adapted to the

specific context of Central American and Dominican Republic countries. Additional modifications were made based on input from personnel in national vector surveillance and control programs.

The questionnaire collects information on the following domains:

- Situational analysis of mosquito-borne diseases.
- Essential programmatic components include governance, financing, human resources, infrastructure, logistics, and information systems.
- Vector surveillance and control activities for *Anopheles*, *Aedes*, and other mosquito-relevant vector species.
- Status of community participation and risk communication efforts.
- Cross-country communication and efforts.

The instrument will be administered in either Spanish or English, according to the respondent's language preference.

C. Sampling Methodology

1. Sampling and Participant Identification

The assessment targets stakeholders at multiple administrative levels, including ministries and secretariats of health (public health entomology laboratories, vector control programs, academic institutions involved in medical entomology, etc.), as well as any other ministries involved in the strategic efforts, and implementing partners such as NGOs and international cooperation agencies. Key stakeholders will be identified through a key stakeholder mapping exercise—drawing on networks such as EntoNet and the SE-COMISCA, Technical Group on Entomology, and others on reference—and invited to participate. This sampling ensures that the assessment reflects both epidemiological priorities and programmatic implementation realities, while capturing the diversity of regional contexts, holistic views, and representativeness.

The assessment will be implemented in two sequential phases. Different communication methods will be implemented (in-person, virtual meetings, phone calls, etc.) and adapted to the situation to reach key participants.

a) Phase 1 – National-Level Assessment (Preparatory Stage)

This initial phase will involve key stakeholders at the central level of the Ministries and Secretariats of Health in the eight participating countries. A census approach will be applied to ensure that relevant national actors involved for vector-borne diseases are included. This includes national program directors, epidemiologists responsible for mosquito-borne diseases, public health entomologists, and laboratory coordinators (where applicable). The goal is to capture a comprehensive understanding of strategic planning, governance, financing, and normative frameworks guiding vector surveillance and control.

As part of the preparatory stage, a pilot assessment will be conducted in Guatemala at the national level, given its role as the headquarters of EntoNet. To test and refine the assessment tools and procedures, a small purposive sample of 3 to 5 participants will be selected. These individuals will hold relevant technical and/or strategic roles and will be drawn from the broader pool of professionals involved in vector surveillance and control. However, they will not be selected from the most critical or high-priority areas to preserve the integrity of the main sample for Phase 1 and avoid overburdening key national experts.

The pilot will serve to validate the methodology and fine-tune the instruments before full implementation. Once complete, data collection will proceed to the national-level assessment under Phase 1, followed by the subnational and local-level assessment in Phase 2.

Phase 2 – Subnational-Level Assessment

This phase will focus on subnational levels, including both departmental (second administrative level) and municipal/local levels (third administrative level) according to the administrative divisions in each country.

Subnational areas will be selected through a non-probabilistic, purposive sampling strategy informed by expert recommendations, commonly referred to as snowball sampling or *key informant referral sampling* (Pahwa, Cavanagh, and Vanstone 2023). At least 30% of interviews conducted in the context will be surveyed to reach a sufficient sample size to reach saturation of views/opinions on the topic (Guest, Bunce and Johnson 2006). National-level stakeholders will assist in identifying priority regions based on:

- Areas with the highest *Aedes* vectored diseases including dengue, ZIKA and Chikungunya,
- Zones that implement at least 30% of the national vector surveillance and control strategy
- Border regions between countries with active transmission
- Areas with large population centres or dense urban settings prioritizing areas with high incidence of mosquito-borne diseases covering urban, peri-urban, and rural contexts
- Operationally relevant subnational programs with entomological activities

At each selected subnational site, participants will include departmental or municipal-level vector control coordinators, entomology field staff, and other relevant public health personnel as needed during the sampling process.

D. Data Collection

Data will be collected through guided interviews and/or targeted consultations with national and regional participants, having in-person meetings. For cases where in-person interviews are not feasible, the assessment will be administered as an online

meeting, a phone call, or an online questionnaire using REDCap. REDCap is a secure, web-based platform that supports real-time data entry, automated validation, and multi-language interfaces, ensuring accurate and efficient data capture from field teams.

National and regional focal points, in collaboration with SE-COMISCA and participating Ministries of Health, will coordinate logistics and follow-up to ensure high response rates as well as time provided to answer the questionnaire. REDCap questionnaire will require for all questions to be answered to minimize left-out or blank responses. In instances of inconsistent responses and upon review, researchers will follow up directly with participants to validate or clarify the information—provided that participants indicated their willingness to be contacted by email or phone after submitting the questionnaire. Even though the questionnaire asks for name, institution, and country (for internal control purposes), all responses will be coded anonymized before analysis to protect the confidentiality of participants and institutions.

To ensure confidentiality while allowing for internal data management and tracking, each interview will be assigned a unique anonymized code. The code structure follows the format [COUNTRY]_[PHASE]_[ID], where COUNTRY is the two-letter ISO code (e.g., GT for Guatemala), PHASE indicates whether the interview was conducted during Phase 1 (national level, coded as F1) or Phase 2 (subnational level, coded as F2), and ID is a sequential number assigned per country and phase. For example, the code *HN_F2_004* refers to the fourth interview conducted in Honduras during Phase 2. A confidential internal reference table with these codes and participants' names and institutions will be maintained securely by the core research team for data validation purposes only. No personal or institutional identifiers will be included in the analysis dataset. No personal or institutional identifiers will be included in the analysis dataset.

E. Data Analysis

Collected data will be analysed using both descriptive and categorical methods. Categorical variables will be summarized as frequencies and percentages. Responses to the needs assessment items will be classified into the following four categories and their numerical equivalent according to the answers for further analytical measures, adapted from van den Berg et al. (2024):

- a) Adequate (no improvement needed)
- b) Partially adequate (component present but requires updating or improvement)
- c) Inadequate or absent (clear need for improvement), and
- d) No information provided (missing data),

Numerical variables will be analysed using measures of central tendency and dispersion. Data cleaning, transformation, and analysis will be performed using RStudio (Posit Team, 2023; R Core Team, 2022). Descriptive statistics will be generated with the “summary” package (Sjoberg et al., 2021), and data visualization will be conducted using the “ggplot2” package (Wickham, 2016).

F. Ethical Considerations

Prior to the initiation of the assessment, institutional approval will be obtained from the Research Ethics Committee of the Universidad del Valle de Guatemala. This assessment does not constitute human subjects research, as it is focused on evaluating system-level capacities, resources, and gaps, rather than generating new knowledge or testing hypotheses.

Nonetheless, ethical safeguards will be rigorously applied. All participants will receive clear information outlining the assessment's objectives, scope, and intended use of data. Informed consent (see annex 1) will be obtained before participation. To ensure confidentiality, all individual-level data will be handled securely, and findings will be reported exclusively in aggregated form, preventing identification of individuals or institutions.

G. Data Management

All data will be stored on secure, encrypted servers with restricted access, in compliance with international data protection and cybersecurity standards. Public dissemination of country-specific data will occur only with written authorization from the respective Ministries of Health, in full respect of national sovereignty over strategic public health information.

Two copies of the database will be generated. The final dataset will be owned by the Ministry of Health of Guatemala, while authorized collaborating institutions—UVG and the CDC—will have access for purposes of data analysis. For the shared databases, the only identifiers will be GPS coordinates of surveyed locations and codes (See section D. Data Collection); no names, addresses, or other personal identifiers will be shared.

H. Dissemination of Results

To ensure that findings lead to tangible action, results will be disseminated through tailored outputs targeted at different stakeholder groups. A regional technical report will present cross-country analyses, highlighting shared challenges, regional trends, and opportunities for cross-border collaboration. In addition, country-specific reports will be developed to provide detailed diagnostics and actionable recommendations aligned with each national context.

Aggregated data will be made publicly available through the EntoNet Data Hub, an open-access regional repository, with username and password access. In contrast, country-level data will remain confidential and will only be shared with third parties upon explicit authorization from the respective Ministries of Health

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V. List of Annexes

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